

JPE Replication Report

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1 Paper Identification

ID	20250178
Author	Yang
Title	Appropriate Entrepreneurship? The Rise of China and the Developing World
Iteration	1

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. Please cite all data in readme and paper, also the confidential one.
2. On the PitchBook access conditions, please give at least an indicative price range to obtain the data.
3. There can only be one unique README in your deposit, so I am asking you to copy what is currently in `code/BERT_generation` into your readme. Delete the other one. Also paste `provenance.md` into the readme. Explain clearly your exact settings for the BERT runs.
4. It looks to me like your `requirements.txt` are not sufficient to run your BERT model. There is no `pytorch`. I need your full, exhaustive software and hardware stack where you generate those BERT outputs. You can put a second `requirements.txt` into the BERT folder and explain in the (main, and only) README how I would go about running this again. When I say full compute stack then I mean the precise spec of CPU/GPU, precise BERT version number, etc
5. We hit a type error at some point. What's wrong there?

3 General

- ☒ Data Availability and Provenance Statements
 - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☒ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☒ License for Data (optional, but recommended)
 - ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements
 - ☒ Software Requirements
 - ☒ Controlled Randomness (as necessary)
 - ☒ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☒ License for Code (Optional, but recommended)
- ☒ {REPLICATOR} to Replicators
 - ☒ Details (as necessary)
- ☒ List of tables and programs
- ☒ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.

- ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
- ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

{REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

{REQUIRED} Please provide affirmation of support for replication checks. – As per the [JPE policy](#), “In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication.”

{REQUIRED} Please provide a codebook for the data. – The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

5.1 Data Sources

file/name	public	private	DOI/URL	Access describe
suitability_resource/WDI_full.xlsx	yes	no	https://databank.worldbank.org/source/world-development-indicators (CC-BY 4.0)	no
patent_resource/g_patent.tsv	yes	no	https://patentsview.org/download/data-download-tables (CC-BY 4.0)	yes
patent_resource/g_us_patent_citation.tsv	yes	no	https://patentsview.org/download/data-download-tables (CC-BY 4.0)	yes
patent_resource/g_cpc_current.tsv	yes	no	https://patentsview.org/download/data-download-tables (CC-BY 4.0)	yes
patent_resource/g_application.tsv	yes	no	https://patentsview.org/download/data-download-tables (CC-BY 4.0)	yes
patent_resource/g_assignee_disambiguated.tsv	yes	no	https://patentsview.org/download/data-download-tables (CC-BY 4.0)	yes
patent_resource/g_location_disambiguated.tsv	yes	no	https://patentsview.org/download/data-download-tables (CC-BY 4.0)	yes
patent_resource/g_inventor_disambiguated.tsv	yes	no	https://patentsview.org/download/data-download-tables (CC-BY 4.0)	yes
geocoding_resource/ne_10m_admin_0_countries/ (shapefile set)	yes	no	https://www.naturalearthdata.com (Admin 0–Countries; Populated Places. 1:10m Cultural Vectors)	no
geocoding_resource/ne_10m_populated_places. {shp,dbf,shx,prj,cpg}	yes	no	https://www.naturalearthdata.com (Admin 0–Countries; Populated Places. 1:10m Cultural Vectors)	no
geocoding_resource/simplemaps.csv	yes	no	https://simplemaps.com/data/world-cities (CC-BY 4.0)	no
geocoding_resource/opedatasoft.csv	yes	no	https://public.opendatasoft.com/explore/dataset/geonames-all-cities-with-a-population-1000/ (CC-BY 4.0)	no
other_data_resource/un_polity_china.dta (+ twin)	yes	no	https://doi.org/10.7910/DVN/LEJUQZ	no
other_data_resource/Trade_Volume_Export_I mport.csv	yes	no	https://comtradeplus.un.org.	no

file/name	public	private	DOI/URL	Access describe
other_data_resource/table1_panelB_funding_sources.csv	yes	no	Not provided	no
pitchbook/pbdeals.dta	no	yes	Not provided	yes
pitchbook/pbcompanies.dta	no	yes	Not provided	yes
pitchbook/pbinvestors.dta	no	yes	Not provided	yes
pitchbook/pbdealinvestor.dta	no	yes	Not provided	yes
pitchbook/pbmarketmap_v2.csv	no	yes	Not provided	yes
pitchbook/pbdealinvestor.dta	no	yes	Not provided	yes
pitchbook/later_years_extension/All_VC_Comp_Deal_Investor_2022-2024.csv	no	yes	Not provided	yes
pitchbook/later_years_extension/pbmarketmap_22_24.csv	no	yes	Not provided	yes
other_data_resource/LP.dta	no	yes	Not provided	yes

6 Requirements

- ✓ README is in TXT, MD, or PDF format
- ✓ README is accessible at the root of the package (not inside a folder)
- ✓ Title conforms to guidance (starts with “Data and Code for: Title of Paper” or “Code for: Title of Paper”, is properly capitalized)
- ✓ Authors (with affiliations) are listed in the same order as on the paper

6.1 Stated Requirements

- ☐ No requirements specified
- ✓ Software Requirements specified as follows:
 - Stata 19.5
 - Python >= 3.11
- ✓ Computational Requirements specified as follows:
 - At least 64gb of memory, parallel computing capabilities.
- ✓ Time Requirements specified as follows:
 - If parallel computing: ~8-10 hours. If not, could be up to 48 hours.
- ✓ Requirements are complete.

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals. If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this – please check!

Summary: – Data files with PII indicators: 58 – Variables flagged in data: 437 – Code files with PII references: 113 – PII references in code: 5345

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	All_VC_Comp_Deal_Investor_2022–2024.csv	35	name, lname, loc, location, address, city, country, phone, fax, email, lat, url, social
Data	CountryDataset.2024019.supplemented.xlsx	61	country, name, lat
Data	EM_policy_constraints.csv	2	name, country

File Type	File	Variables/References	PII Categories
Data	EM_policy_constraints.dta	2	name, country
Data	LP.csv	4	name, loc, location, city
Data	LP.dta	4	name, loc, location, city
Data	OECD_list.csv	1	country
Data	OECD_list.dta	1	country
Data	WDI_full.xlsx	4	country, name
Data	company_level_predictions_v2.csv	13	lname, name, country
Data	company_level_predictions_v2.dta	13	lname, name, country
Data	country_crosswalk.csv	2	country
Data	country_crosswalk.dta	2	country
Data	country_panel.csv	1	country
Data	country_panel.dta	1	country
Data	critical_subseg.csv	5	network, son, lname, name
Data	critical_subseg.dta	5	network, son, lname, name
Data	crosswalk.xlsx	2	country, lname, name
Data	figure1c_IP0_CapitalIQ_to_PB_EM_0901_Final.xlsx	24	name, country
Data	figure1c_IP0_CapitalIQ_to_PB_US_Final.xlsx	5	name, country
Data	figure1c_IP0_CapitalIQ_to_PB_nonEM_nonUS.With_JL_Additions.xlsx	38	name, country
Data	indicators_WDI_MtV_corrected.xlsx	2	name, block, loc
Data	indicators_WDI_MtV_from_existing_no_trade2.xlsx	2	name, block, loc
Data	indicators_WDI_MtV_from_scratch_allassigned.xlsx	2	name, block, loc
Data	indicators_WDI_MtV_gpt_assigned.xlsx	2	name, block, loc
Data	ne_10m_admin_0_countries.dbf	34	name, lon
Data	ne_10m_populated_places.dbf	40	name, city, lat, lon
Data	opendatasoft.csv	10	name, country, lat, coord
Data	patent_countrysector_similarity_2000_2013_100k_new_china.csv	2	country
Data	patent_countrysector_similarity_2000_2013_100k_new_china.dta	2	country
Data	patent_sector_labels_grantyear.csv	1	country
Data	patent_sector_labels_grantyear.dta	1	country
Data	patent_sector_labels_new_grantyear.csv	1	country
Data	patent_sector_labels_new_grantyear.dta	1	country
Data	pbcompanies.csv	18	name, lname, loc, location, address, city, country, phone, fax, email, url, social

File Type	File	Variables/References	PII Categories
Data	pbcompanies.dta	18	name, lname, loc, location, address, city, country, phone, fax, email, url, social
Data	pbdealinvestor.csv	3	name
Data	pbdealinvestor.dta	3	name
Data	pbdeals.csv	5	name, lat, loc, location
Data	pbdeals.dta	5	name, lat, loc, location
Data	pbinvestors.csv	12	name, lname, loc, location, address, city, country, phone, fax, email
Data	pbinvestors.dta	12	name, lname, loc, location, address, city, country, phone, fax, email
Data	pbmarketmap_22_24.csv	1	lname, name
Data	pbmarketmap_v2.csv	1	name
Data	policy_constrained_sectors.csv	1	name
Data	policy_constrained_sectors.dta	2	name, lname
Data	similarity_at_sector_x_country_x_year.csv	3	lname, name, country, lat
Data	similarity_at_sector_x_country_x_year.dta	3	lname, name, country, lat
Data	similarity_compiled.csv	2	name, lname
Data	similarity_compiled_western.csv	1	lname, name
Data	simplemaps.csv	6	city, lat, country, name
Data	table1_panelC_Copy of VC Returns 2024Q4 v2.xlsx	1	street
Data	tableA1_VCpc.xlsx	1	country
Data	tableA1_published_values.csv	1	country
Data	un_polity_china.csv	1	country
Data	un_polity_china.dta	2	country
Data	worldwide_pairs_filtered_no_EPO_countries.csv	5	country, lname, name
Data	worldwide_pairs_filtered_no_EPO_countries.dta	5	country, lname, name
Code	00_run_all.py	4	name
Code	01_generate_analysis_v2.do	97	loc, lat, country, name, lname, lon, location
Code	01_prepare_manifest.py	1	name
Code	02_generate_suitability.py	83	country, loc, name, zip, lat, lon, block
Code	02_train_bert_categories.py	16	son, lat, lname, name
Code	03_predict_companies.py	7	lname, name
Code	03a_generate_populated_places.do	9	loc, lat, city, lon, name
Code	03b_generate_controls.py	20	country, loc, name, lon, phone, lat

File Type	File	Variables/References	PII Categories
Code	03c_generate_regression_auxiliary.do	133	loc, country, lname, name, second, lat, location, lon
Code	04_generate_regression_corrected_120623.do	40	loc, lat, country, lname, name, block
Code	04_make_predicted_positive.py	14	name, lname
Code	05_make_company_level_predictions.py	13	country, name, lname
Code	05a_generate_postregression_inputs.do	22	loc, country, name, block
Code	05b_generate_suitability_gdp_component.do	9	loc, country
Code	05c_generate_regression_alltype_by_cat.do	63	loc, second, lat, country, name, lname, lon, block
Code	05d_generate_dealcount_alltype_cluster.do	30	loc, lname, name, country, lon
Code	05e_generate_pre_period_deals.do	7	loc, lname, name, country
Code	05f_generate_similarity_western.do	21	loc, name, lname, country
Code	05g_generate_regression_validation_countrypair.do	88	loc, country, name, lat, lon, lname
Code	05h_generate_deal_22_24.do	67	loc, lat, name, lname, country, lon
Code	05i_generate_regression_00_24_integrated.do	17	loc, country, lname, name, block
Code	06a_generate_patent_citations_from_raw.py	22	loc, location, country, name
Code	06b_generate_patent_layers.py	48	country, lname, name, loc
Code	06c_supplement_regression_patents.do	8	loc, lname, name, country
Code	07b_generate_patent_geolocation_from_raw.py	22	loc, location, city, country, lat, lon, name
Code	07c_generate_company_geolocation.py	58	country, name, lname, loc, city, location, lat, coord, lon
Code	07d_generate_analysis_city.do	131	loc, lat, name, city, lon, location, country, lname
Code	08a_generate_regression_a13_variants.py	4	name, country
Code	08b_generate_regression_figureA11_dropped.do	28	loc, name, block, country, lon, lat
Code	09_generate_simulated_deals.do	142	loc, lat, lon, country, name, lname, block
Code	09_generate_simulated_deals.sh	9	lat, name
Code	bert_pipeline_lib.py	32	name, lname, block, loc, son
Code	figure_1.py	44	lat, block, loc, son, zip, name

File Type	File	Variables/References	PII Categories
Code	<code>figure_2.do</code>	3	loc
Code	<code>figure_3.do</code>	26	loc, name, country, lat, block
Code	<code>figure_4.do</code>	3	loc
Code	<code>figure_A1.py</code>	30	name, country, loc
Code	<code>figure_A10.py</code>	14	name, country, zip, loc
Code	<code>figure_A11.do</code>	7	loc
Code	<code>figure_A12.do</code>	17	loc, country, lname, name
Code	<code>figure_A13.do</code>	3	loc
Code	<code>figure_A14.do</code>	11	loc
Code	<code>figure_A15.do</code>	7	loc
Code	<code>figure_A16.do</code>	3	loc
Code	<code>figure_A17.do</code>	32	loc, city, lat, country, name
Code	<code>figure_A2.do</code>	11	loc, country, name
Code	<code>figure_A3.do</code>	5	loc, name, lat
Code	<code>figure_A4.py</code>	6	name, son, loc
Code	<code>figure_A5.py</code>	7	name, second, lat, lname
Code	<code>figure_A6.do</code>	18	loc, country, name, lat
Code	<code>figure_A7.py</code>	16	lat, lon, name, country, block, loc
Code	<code>figure_A8.py</code>	5	name, country, lat
Code	<code>figure_A9.py</code>	16	name, country, lat, loc
Code	<code>generate_all.sh</code>	8	name, lat, country, loc, location, city
Code	<code>install.do</code>	5	loc
Code	<code>make_patent_sector_labels.py</code>	25	name, country, loc, lat, location
Code	<code>patent_additional_prediction_100k_china_win.ipynb</code>	63	lat, name, country, loc, location, son, block, second, network, house
Code	<code>patent_additional_prediction_win.ipynb</code>	90	lat, name, son, block, loc, second, network, house
Code	<code>patent_family_exercise.ipynb</code>	167	lat, name, loc, location, country, lname, block
Code	<code>patent_prediction_analysis.ipynb</code>	33	lat, son, loc, location, country, name, lname
Code	<code>patent_prediction_analysis_new100k.ipynb</code>	29	lat, loc, location, name, country, lname
Code	<code>patent_similarity.ipynb</code>	157	lat, country, name, loc, son, block, second, house, network, lname

File Type	File	Variables/References	PII Categories
Code	patent_similarity.py	14	country, lat, name
Code	paths.do	7	loc, lat
Code	predict_only.py	14	lat, name, lname
Code	predict_patent_sectors.py	33	loc, location, lat, name, country
Code	run_all.sh	2	name, lat
Code	similarity_clean.ipynb	70	lat, country, name, loc, lname, second
Code	similarity_clean.py	27	name, country, loc, lname, lat
Code	similarity_clean_detail_western.py	30	lat, country, name, loc, lname
Code	similarity_compile_2025.ipynb	855	name, block, loc, son, lat, house, network, second, lon, lname, country
Code	similarity_mac.ipynb	278	lat, lname, name, house, block, loc, second
Code	slurm_predict_patents_batch_01.sh	5	name, url, block, loc
Code	table_1.py	22	son, country, name, street
Code	table_2.do	29	loc, country
Code	table_3.do	26	loc, country, name
Code	table_4.do	49	loc, name, country
Code	table_5.do	10	loc, country
Code	table_6.do	35	loc, name, country
Code	table_7.do	24	loc, country, name
Code	table_8.do	33	loc, country, name
Code	table_A1.py	41	name, country, loc
Code	table_A10.do	15	loc, name, country
Code	table_A11.do	12	loc, country, block, lname, name
Code	table_A12.do	31	loc, country, block, lname, name
Code	table_A13.do	15	loc, country
Code	table_A14.do	105	loc, block, country, name
Code	table_A15.do	35	loc, phone, country
Code	table_A16.do	19	loc, country
Code	table_A17.do	16	loc, country
Code	table_A18.do	40	loc, country
Code	table_A2.do	85	loc, name, country, lname
Code	table_A20.do	20	loc, country

File Type	File	Variables/References	PII Categories
Code	table_A21.do	15	loc, country, name
Code	table_A22.do	24	loc, country
Code	table_A23.do	22	loc, country
Code	table_A24.do	22	loc, country
Code	table_A25.do	16	loc, country
Code	table_A26.do	16	loc, country
Code	table_A27.do	8	loc, country
Code	table_A28.do	12	loc, country
Code	table_A29.do	51	loc, city, lat, country, name
Code	table_A30.do	57	loc, city, lat, country, name
Code	table_A4.do	20	loc, country, lat, name, lon, lname
Code	table_A5.do	22	loc, country
Code	table_A6.do	18	loc, lname, name, country
Code	table_A7.do	19	loc, country
Code	table_A8.do	27	loc, lat, country
Code	table_A9.do	12	loc, country
Code	train_predict_crossvalidate_mac.py	13	name, lname, loc, lat
Code	train_predict_pat01.ipynb	26	loc, location, country, lname, name, son
Code	train_predict_patent_full.ipynb	804	loc, lname, name, lat, son, second
Code	train_predict_v2-1.ipynb	18	name, lname, loc, lat

See [Appendix](#) for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-07-15T12:41:59.474

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any `windows` filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that `STATA` allows `/` to be a filepath separator on a windows platform, hence this is a requirement for *all* `STATA` applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (<code>C:\</code> etc)
230	12	76	7	2

7.1.3 Duplicate Files Report

We found the following duplicate files:

Filename	Size (MB)	Checksum (MD5)
/JPE_replication_for_submission/data/raw/geocoding_resource/ne_10m_admin_0_countries/ne_10m_admin_0_countries.cpg	0.0	663b90c899fa25a111067b
/__MACOSX/JPE_replication_for_submission/code/._.DS_Store	0.0	0bed7e90c2bade9763fa18f
/__MACOSX/JPE_replication_for_submission/code/BERT_generation/._.DS_Store	0.0	0bed7e90c2bade9763fa18f
/__MACOSX/JPE_replication_for_submission/confidential-data-not-for-publication/._.DS_Store	0.0	0bed7e90c2bade9763fa18f
/__MACOSX/JPE_replication_for_submission/confidential-data-not-for-publication/Analysis/._.DS_Store	0.0	0bed7e90c2bade9763fa18f
/__MACOSX/JPE_replication_for_submission/confidential-data-not-for-publication/Raw/._.DS_Store	0.0	0bed7e90c2bade9763fa18f
/__MACOSX/JPE_replication_for_submission/confidential-data-not-for-publication/Raw/other_data_resource/._.DS_Store	0.0	0bed7e90c2bade9763fa18f
/__MACOSX/JPE_replication_for_submission/data/._.DS_Store	0.0	0bed7e90c2bade9763fa18f
/__MACOSX/JPE_replication_for_submission/data/raw/._.DS_Store	0.0	0bed7e90c2bade9763fa18f
/__MACOSX/JPE_replication_for_submission/data/raw/other_data_resource/._.DS_Store	0.0	0bed7e90c2bade9763fa18f
/__MACOSX/JPE_replication_for_submission/output/._.DS_Store	0.0	0bed7e90c2bade9763fa18f
/__MACOSX/JPE_replication_for_submission/output/logs/._.DS_Store	0.0	0bed7e90c2bade9763fa18f

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We found the following files larger than 100MB:

Filename	Size (MB)
/JPE_replication_for_submission/confidential-data-not-for-publication/Raw/BERT_prediction_resource/company_level_predictions_v2.csv	130.34
/JPE_replication_for_submission/confidential-data-not-for-publication/Raw/pitchbook/pbinvestors.csv	159.48
/JPE_replication_for_submission/confidential-data-not-for-publication/Raw/pitchbook/pbdealinvestor.csv	168.56
/JPE_replication_for_submission/data/raw/patent_resource/g_application.tsv	378.92
/JPE_replication_for_submission/confidential-data-not-for-publication/Raw/pitchbook/pbcompanies.csv	552.9
/JPE_replication_for_submission/confidential-data-not-for-publication/Raw/pitchbook/pbdealinvestor.dta	828.59
/JPE_replication_for_submission/data/raw/patent_resource/g_assignee_disambiguated.tsv	946.37
/JPE_replication_for_submission/confidential-data-not-for-publication/Raw/pitchbook/pbdeals.csv	1007.18
/JPE_replication_for_submission/confidential-data-not-for-publication/Raw/BERT_prediction_resource/company_level_predictions_v2.dta	1065.36
/JPE_replication_for_submission/data/raw/patent_resource/g_inventor_disambiguated.tsv	1908.57
/JPE_replication_for_submission/confidential-data-not-for-publication/Raw/pitchbook/pbinvestors.dta	2184.39
/JPE_replication_for_submission/data/raw/patent_resource/g_cpc_current.tsv	2749.25
/JPE_replication_for_submission/confidential-data-not-for-publication/Raw/pitchbook/pbcompanies.dta	4570.82
/JPE_replication_for_submission/confidential-data-not-for-publication/Raw/pitchbook/later_years_extension/All_VC_Comp_Deal_Investor_2022-	4989.19

Filename	Size (MB)
2024.csv	
/JPE_replication_for_submission/data/raw/patent_resource/g_patent.tsv	6203.5
/JPE_replication_for_submission/confidential-data-not-for-publication/Raw/pitchbook/pbdeals.dta	8674.91
/JPE_replication_for_submission/data/raw/patent_resource/g_us_patent_citation.tsv	9156.58

7.2 Code description

- Stage A prepares data.
- Each sub script in stage B generates one exhibit.

cloc github.com/AIDanial/cloc v 2.02 T=3.49 s (43.0 files/s, 708166.1 lines/s)				
Language	files	blank	comment	code
CSV	24	0	0	2440589
Stata	66	1223	9	7610
Python	33	716	149	3817
Jupyter Notebook	12	0	13590	1747
HTML	2	220	52	732
Markdown	3	76	0	384
Bourne Shell	4	41	10	318
Text	6	9	0	95
---	---	---	---	---
SUM:	150	2285	13810	2455292

☒ The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

NOTE: In-text numbers that reference numbers in tables do not need to be listed. Only in-text numbers that correspond to no table or figure need to be listed.

7.3 Computing Environment of the Replicator

- Onyxia [RStudio](#) instance with 25000m CPU allocation and 124 Gb of memory, 100Gb of persistent storage. Custom Docker image ([see here](#)) based on [dataeditors/stata19_5-mp](#)
- Running Debian 6.12.86-1 on [Intel\(R\) Xeon\(R\) Gold 6448Y](#) with 32 cores.
- Stata/MP 19.5
- Pyton 3.12 in a clean virtual environment

8 Replication

- Downloaded code and data from dropbox.
- Committed code to git repo.
- Used a custom [stata19_5-mp](#) Docker image to run on the instance.
- Installed python's [venv](#) package and activated a new virtual environment at the root of the filesystem in Onyxia service (as per [README](#) recommendations)
- Stan ran [code/run_all.sh](#)
- Stage A ran without any issues (step 1 through 9)
- Some steps in Stage B did not succeed:

- Two Stage B steps were skipped:

```
>>> SKIPPED: code/tables/table_A8.do (missing input confidential-data-not-for-publication/Analysis/intermediate_and_other_data/simulated_
>>> SKIPPED: code/figures/figure_A11.do (missing input confidential-data-not-for-publication/Analysis/intermediate_and_other_data/regress
```

However, stage A output (which is in terminal, so perhaps lost error messages?) explicitly reports no errors, and all sub-scripts were run.

- One Stage B step failed due to a `TypeError`:

```
>>> python: code/figures/figure_A9.py
/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/text.py:905: FutureWarning: Calling float on a single element Series
x = float(self.convert_xunits(self._x))
Traceback (most recent call last):
  File "/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/artist.py", line 1402, in _get_tightbbox_for_layout_only
    return obj.get_tightbbox(*args, **kwargs, "for_layout_only": True))
  File "/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/axes/_base.py", line 4542, in get_tightbbox
    bbox = a.get_tightbbox(renderer)
  File "/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/artist.py", line 364, in get_tightbbox
    bbox = self.get_window_extent(renderer)
  File "/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/text.py", line 970, in get_window_extent
    x, y = self.get_unitless_position()
  File "/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/text.py", line 906, in get_unitless_position
    y = float(self.convert_yunits(self._y))
TypeError: only 0-dimensional arrays can be converted to Python scalars

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
  File "/home/onyxia/work/code/figures/figure_A9.py", line 105, in <module>
    main()
  File "/home/onyxia/work/code/figures/figure_A9.py", line 102, in main
    plot_sector(df, seg, out_name)
  File "/home/onyxia/work/code/figures/figure_A9.py", line 91, in plot_sector
    plt.tight_layout()
  File "/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/pyplot.py", line 2835, in tight_layout
    gcf().tight_layout(pad=pad, h_pad=h_pad, w_pad=w_pad, rect=rect)
  File "/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/figure.py", line 3640, in tight_layout
    engine.execute(self)
  File "/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/layout_engine.py", line 183, in execute
    kwargs = get_tight_layout_figure(
  File "/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/_tight_layout.py", line 266, in get_tight_layout_figure
    kwargs = _auto_adjust_subplotpars(fig, renderer,
  File "/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/_tight_layout.py", line 82, in _auto_adjust_subplotpars
    bb += [matplotlib.artist.get_tightbbox_for_layout_only(ax, renderer)]
  File "/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/artist.py", line 1404, in _get_tightbbox_for_layout_only
    return obj.get_tightbbox(*args, **kwargs)
  File "/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/axes/_base.py", line 4542, in get_tightbbox
    bbox = a.get_tightbbox(renderer)
  File "/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/artist.py", line 364, in get_tightbbox
    bbox = self.get_window_extent(renderer)
  File "/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/text.py", line 970, in get_window_extent
    x, y = self.get_unitless_position()
  File "/home/onyxia/work/venv_yang/lib/python3.12/site-packages/matplotlib/text.py", line 906, in get_unitless_position
    y = float(self.convert_yunits(self._y))
TypeError: only 0-dimensional arrays can be converted to Python scalars
ERROR: code/figures/figure_A9.py failed
```

- All other exhibits were generated.

9 Findings

9.1 Data Preparation Code

No errors reported for data preparation code (Stage A), but 2 scripts were skipped (see section 8 'Replication' above for more details).

9.2 Tables and Figures

Data Preparation Program	Dataset created	Reproduced?	
Files in code/BERT_generation	9 files in BERT_prediction_resource/ (+ CSV twins)	Yes	
Figure/Table #	Program	Line Number	Reproduced?
Table 1	code/tables/table_1.py	-	Yes
Table 2	code/tables/table_2.do	-	Yes
Table 3	code/tables/table_3.do	-	Yes
Table 4	code/tables/table_4.do	-	Yes
Table 5	code/tables/table_5.do	-	Yes
Table 6	code/tables/table_6.do	-	Yes
Table 7	code/tables/table_7.do	-	Yes
Table 8	code/tables/table_8.do	-	Yes
Table A1	code/tables/table_A1.py	-	Yes
Tab e A3	No script (textual data)	-	-
Table A2	code/tables/table_A2.do	-	Yes
Table A4	code/tables/table_A4.do	-	Yes
Table A5	code/tables/table_A5.do	-	Yes
Table A6	code/tables/table_A6.do	-	Yes
Table A7	code/tables/table_A7.do	-	Yes
Table A8	code/tables/table_A8.do	-	No
Table A9	code/tables/table_A9.do	-	Yes
Table A10	code/tables/table_A10.do	-	Yes
Table A11	code/tables/table_A11.do	-	Yes
Table A12	code/tables/table_A12.do	-	Yes
Table A13	code/tables/table_A13.do	-	Yes
Table A14	code/tables/table_A14.do	-	Yes
Table A15	code/tables/table_A15.do	-	Yes
Table A16	code/tables/table_A16.do	-	Yes
Table A17	code/tables/table_A17.do	-	Yes
Table A18	code/tables/table_A18.do	-	Yes
Table A19	No script (textual data)	-	-
Table A20	code/tables/table_A20.do	-	Yes
Table A21	code/tables/table_A21.do	-	Yes
Table A22	code/tables/table_A22.do	-	Yes
Table A23	code/tables/table_A23.do	-	Yes

Data Preparation Program	Dataset created	Reproduced?	
Table A24	code/tables/table_A24.do	-	Yes
Table A25	code/tables/table_A25.do	-	Yes
Table A26	code/tables/table_A26.do	-	Yes
Table A27	code/tables/table_A27.do	-	Yes
Table A28	code/tables/table_A28.do	-	Yes
Table A29	code/tables/table_A29.do	-	Yes
Table A30	code/tables/table_A30.do	-	Yes
Figure 1	code/figures/figure_1.py	-	Yes
Figure 2	code/figures/figure_2.do	-	Yes
Figure 3	code/figures/figure_3.do	-	Yes
Figure 4	code/figures/figure_4.do	-	Yes
Figure A1	code/figures/figure_A1.do	-	Yes
Figure A2	code/figures/figure_A2.do	-	Yes
Figure A3	code/figures/figure_A3.do	-	Yes
Figure A4	code/figures/figure_A4.do	-	Yes
Figure A5	code/figures/figure_A5.do	-	Yes
Figure A6	code/figures/figure_A6.do	-	Yes
Figure A7	code/figures/figure_A7.do	-	Yes
Figure A8	code/figures/figure_A8.do	-	No (skipped)
Figure A9	code/figures/figure_A9.do	-	No
Figure A10	code/figures/figure_A10.do	-	Yes
Figure A11	code/figures/figure_A11.do	-	No (skipped)
Figure A12	code/figures/figure_A12.do	-	Yes
Figure A13	code/figures/figure_A13.do	-	Yes
Figure A14	code/figures/figure_A14.do	-	No (skipped)
Figure A15	code/figures/figure_A15.do	-	No (skipped)
Figure A16	code/figures/figure_A16.do	-	Yes
Figure A17	code/figures/figure_A17.do	-	Yes
In-text numbers	Program	Line Number	

- Two figures (A14 and A14) were not generated, but the scripts did not show any errors.
- The other exhibits not generated (Figures A8 and A11) were logged as skipped (in stage A).
- One error reported for figure A9 (in stage B).

9.3 In-Text Numbers

- ☐ In-text numbers not verified.
- ☒ There are no in-text numbers, or all in-text numbers stem from tables and figures.
- ☐ There are in-text numbers, but they are not identified in the code

9.4 Other Issues

10 Classification

- ☐ full reproduction
- ☐ full reproduction with minor issues
- ☒ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☐ None.
- ☐ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☐ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check

☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☒ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☒ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements

10.2 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. As a reminder, *privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals*. If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary: - Data files with PII indicators: 58 - Variables flagged in data: 437 - Code files with PII references: 113 - PII references in code: 5345

10.2.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	All_VC_Comp_Deal_Investor_2022-2024.csv	35	name, lname, loc, location, address, city, country, phone, fax, email, lat, url, social
Data	CountryDataset.2024019.supplemented.xlsx	61	country, name, lat
Data	EM_policy_constraints.csv	2	name, country
Data	EM_policy_constraints.dta	2	name, country
Data	LP.csv	4	name, loc, location, city
Data	LP.dta	4	name, loc, location, city
Data	OECD_list.csv	1	country
Data	OECD_list.dta	1	country
Data	WDI_full.xlsx	4	country, name
Data	company_level_predictions_v2.csv	13	lname, name, country
Data	company_level_predictions_v2.dta	13	lname, name, country
Data	country_crosswalk.csv	2	country
Data	country_crosswalk.dta	2	country
Data	country_panel.csv	1	country

File Type	File	Variables/References	PII Categories
Data	country_panel.dta	1	country
Data	critical_subseg.csv	5	network, son, lname, name
Data	critical_subseg.dta	5	network, son, lname, name
Data	crosswalk.xlsx	2	country, lname, name
Data	figure1c_IP0_CapitalIQ_to_PB_EM_0901_Final.xlsx	24	name, country
Data	figure1c_IP0_CapitalIQ_to_PB_US_Final.xlsx	5	name, country
Data	figure1c_IP0_CapitalIQ_to_PB_nonEM_nonUS.With_JL_Additions.xlsx	38	name, country
Data	indicators_WDI_MtV_corrected.xlsx	2	name, block, loc
Data	indicators_WDI_MtV_from_existing_no_trade2.xlsx	2	name, block, loc
Data	indicators_WDI_MtV_from_scratch_allassigned.xlsx	2	name, block, loc
Data	indicators_WDI_MtV_gpt_assigned.xlsx	2	name, block, loc
Data	ne_10m_admin_0_countries.dbf	34	name, lon
Data	ne_10m_populated_places.dbf	40	name, city, lat, lon
Data	opendatasoft.csv	10	name, country, lat, coord
Data	patent_countrysector_similarity_2000_2013_100k_new_china.csv	2	country
Data	patent_countrysector_similarity_2000_2013_100k_new_china.dta	2	country
Data	patent_sector_labels_grantyear.csv	1	country
Data	patent_sector_labels_grantyear.dta	1	country
Data	patent_sector_labels_new_grantyear.csv	1	country
Data	patent_sector_labels_new_grantyear.dta	1	country
Data	pbcompanies.csv	18	name, lname, loc, location, address, city, country, phone, fax, email, url, social
Data	pbcompanies.dta	18	name, lname, loc, location, address, city, country, phone, fax, email, url, social
Data	pbdealinvestor.csv	3	name
Data	pbdealinvestor.dta	3	name
Data	pbdeals.csv	5	name, lat, loc, location
Data	pbdeals.dta	5	name, lat, loc, location
Data	pbinvestors.csv	12	name, lname, loc, location, address, city, country, phone, fax, email
Data	pbinvestors.dta	12	name, lname, loc, location, address, city, country, phone, fax, email
Data	pbmarketmap_22_24.csv	1	lname, name

File Type	File	Variables/References	PII Categories
Data	pbmarketmap_v2.csv	1	name
Data	policy_constrained_sectors.csv	1	name
Data	policy_constrained_sectors.dta	2	name, lname
Data	similarity_at_sector_x_country_x_year.csv	3	lname, name, country, lat
Data	similarity_at_sector_x_country_x_year.dta	3	lname, name, country, lat
Data	similarity_compiled.csv	2	name, lname
Data	similarity_compiled_western.csv	1	lname, name
Data	simplemaps.csv	6	city, lat, country, name
Data	table1_panelC_Copy of VC Returns 2024Q4 v2.xlsx	1	street
Data	tableA1_VCpc.xlsx	1	country
Data	tableA1_published_values.csv	1	country
Data	un_polity_china.csv	1	country
Data	un_polity_china.dta	2	country
Data	worldwide_pairs_filtered_no_EP0_countries.csv	5	country, lname, name
Data	worldwide_pairs_filtered_no_EP0_countries.dta	5	country, lname, name
Code	00_run_all.py	4	name
Code	01_generate_analysis_v2.do	97	loc, lat, country, name, lname, lon, location
Code	01_prepare_manifest.py	1	name
Code	02_generate_suitability.py	83	country, loc, name, zip, lat, lon, block
Code	02_train_bert_categories.py	16	son, lat, lname, name
Code	03_predict_companies.py	7	lname, name
Code	03a_generate_populated_places.do	9	loc, lat, city, lon, name
Code	03b_generate_controls.py	20	country, loc, name, lon, phone, lat
Code	03c_generate_regression_auxiliary.do	133	loc, country, lname, name, second, lat, location, lon
Code	04_generate_regression_corrected_120623.do	40	loc, lat, country, lname, name, block
Code	04_make_predicted_positive.py	14	name, lname
Code	05_make_company_level_predictions.py	13	country, name, lname
Code	05a_generate_postregression_inputs.do	22	loc, country, name, block
Code	05b_generate_suitability_gdp_component.do	9	loc, country
Code	05c_generate_regression_alltype_by_cat.do	63	loc, second, lat, country, name, lname, lon, block
Code	05d_generate_dealcount_alltype_cluster.do	30	loc, lname, name, country, lon
Code	05e_generate_pre_period_deals.do	7	loc, lname, name, country

File Type	File	Variables/References	PII Categories
Code	05f_generate_similarity_western.do	21	loc, name, lname, country
Code	05g_generate_regression_validation_countrypair.do	88	loc, country, name, lat, lon, lname
Code	05h_generate_deal_22_24.do	67	loc, lat, name, lname, country, lon
Code	05i_generate_regression_00_24_integrated.do	17	loc, country, lname, name, block
Code	06a_generate_patent_citations_from_raw.py	22	loc, location, country, name
Code	06b_generate_patent_layers.py	48	country, lname, name, loc
Code	06c_supplement_regression_patents.do	8	loc, lname, name, country
Code	07b_generate_patent_geolocation_from_raw.py	22	loc, location, city, country, lat, lon, name
Code	07c_generate_company_geolocation.py	58	country, name, lname, loc, city, location, lat, coord, lon
Code	07d_generate_analysis_city.do	131	loc, lat, name, city, lon, location, country, lname
Code	08a_generate_regression_a13_variants.py	4	name, country
Code	08b_generate_regression_figureA11_dropped.do	28	loc, name, block, country, lon, lat
Code	09_generate_simulated_deals.do	142	loc, lat, lon, country, name, lname, block
Code	09_generate_simulated_deals.sh	9	lat, name
Code	bert_pipeline_lib.py	32	name, lname, block, loc, son
Code	figure_1.py	44	lat, block, loc, son, zip, name
Code	figure_2.do	3	loc
Code	figure_3.do	26	loc, name, country, lat, block
Code	figure_4.do	3	loc
Code	figure_A1.py	30	name, country, loc
Code	figure_A10.py	14	name, country, zip, loc
Code	figure_A11.do	7	loc
Code	figure_A12.do	17	loc, country, lname, name
Code	figure_A13.do	3	loc
Code	figure_A14.do	11	loc
Code	figure_A15.do	7	loc
Code	figure_A16.do	3	loc

File Type	File	Variables/References	PII Categories
Code	<code>figure_A17.do</code>	32	loc, city, lat, country, name
Code	<code>figure_A2.do</code>	11	loc, country, name
Code	<code>figure_A3.do</code>	5	loc, name, lat
Code	<code>figure_A4.py</code>	6	name, son, loc
Code	<code>figure_A5.py</code>	7	name, second, lat, lname
Code	<code>figure_A6.do</code>	18	loc, country, name, lat
Code	<code>figure_A7.py</code>	16	lat, lon, name, country, block, loc
Code	<code>figure_A8.py</code>	5	name, country, lat
Code	<code>figure_A9.py</code>	16	name, country, lat, loc
Code	<code>generate_all.sh</code>	8	name, lat, country, loc, location, city
Code	<code>install.do</code>	5	loc
Code	<code>make_patent_sector_labels.py</code>	25	name, country, loc, lat, location
Code	<code>patent_additional_prediction_100k_china_win.ipynb</code>	63	lat, name, country, loc, location, son, block, second, network, house
Code	<code>patent_additional_prediction_win.ipynb</code>	90	lat, name, son, block, loc, second, network, house
Code	<code>patent_family_exercise.ipynb</code>	167	lat, name, loc, location, country, lname, block
Code	<code>patent_prediction_analysis.ipynb</code>	33	lat, son, loc, location, country, name, lname
Code	<code>patent_prediction_analysis_new100k.ipynb</code>	29	lat, loc, location, name, country, lname
Code	<code>patent_similarity.ipynb</code>	157	lat, country, name, loc, son, block, second, house, network, lname
Code	<code>patent_similarity.py</code>	14	country, lat, name
Code	<code>paths.do</code>	7	loc, lat
Code	<code>predict_only.py</code>	14	lat, name, lname
Code	<code>predict_patent_sectors.py</code>	33	loc, location, lat, name, country
Code	<code>run_all.sh</code>	2	name, lat
Code	<code>similarity_clean.ipynb</code>	70	lat, country, name, loc, lname, second
Code	<code>similarity_clean.py</code>	27	name, country, loc, lname, lat
Code	<code>similarity_clean_detail_western.py</code>	30	lat, country, name, loc, lname
Code	<code>similarity_compile_2025.ipynb</code>	855	name, block, loc, son, lat, house, network, second, lon, lname, country

File Type	File	Variables/References	PII Categories
Code	<code>similarity_mac.ipynb</code>	278	lat, lname, name, house, block, loc, second
Code	<code>slurm_predict_patents_batch_01.sh</code>	5	name, url, block, loc
Code	<code>table_1.py</code>	22	son, country, name, street
Code	<code>table_2.do</code>	29	loc, country
Code	<code>table_3.do</code>	26	loc, country, name
Code	<code>table_4.do</code>	49	loc, name, country
Code	<code>table_5.do</code>	10	loc, country
Code	<code>table_6.do</code>	35	loc, name, country
Code	<code>table_7.do</code>	24	loc, country, name
Code	<code>table_8.do</code>	33	loc, country, name
Code	<code>table_A1.py</code>	41	name, country, loc
Code	<code>table_A10.do</code>	15	loc, name, country
Code	<code>table_A11.do</code>	12	loc, country, block, lname, name
Code	<code>table_A12.do</code>	31	loc, country, block, lname, name
Code	<code>table_A13.do</code>	15	loc, country
Code	<code>table_A14.do</code>	105	loc, block, country, name
Code	<code>table_A15.do</code>	35	loc, phone, country
Code	<code>table_A16.do</code>	19	loc, country
Code	<code>table_A17.do</code>	16	loc, country
Code	<code>table_A18.do</code>	40	loc, country
Code	<code>table_A2.do</code>	85	loc, name, country, lname
Code	<code>table_A20.do</code>	20	loc, country
Code	<code>table_A21.do</code>	15	loc, country, name
Code	<code>table_A22.do</code>	24	loc, country
Code	<code>table_A23.do</code>	22	loc, country
Code	<code>table_A24.do</code>	22	loc, country
Code	<code>table_A25.do</code>	16	loc, country
Code	<code>table_A26.do</code>	16	loc, country
Code	<code>table_A27.do</code>	8	loc, country
Code	<code>table_A28.do</code>	12	loc, country
Code	<code>table_A29.do</code>	51	loc, city, lat, country, name
Code	<code>table_A30.do</code>	57	loc, city, lat, country, name
Code	<code>table_A4.do</code>	20	loc, country, lat, name, lon, lname

File Type	File	Variables/References	PII Categories
Code	table_A5.do	22	loc, country
Code	table_A6.do	18	loc, lname, name, country
Code	table_A7.do	19	loc, country
Code	table_A8.do	27	loc, lat, country
Code	table_A9.do	12	loc, country
Code	train_predict_crossvalidate_mac.py	13	name, lname, loc, lat
Code	train_predict_pat01.ipynb	26	loc, location, country, lname, name, son
Code	train_predict_patent_full.ipynb	804	loc, lname, name, lat, son, second
Code	train_predict_v2-1.ipynb	18	name, lname, loc, lat

See [Appendix](#) for detailed listing of all flagged instances.